

Week 15 Homework: AI-Generated Images for eCommerce Scenarios

Student: Enis Ogdum

Date: January 24, 2026

Subject: Generative AI Applied in eCommerce - Custom AI Image Generation Experiment

1. Introduction

This report documents an experimental study on the viability and effectiveness of using AI-generated imagery for eCommerce and web service applications. The primary objective was to explore how AI image generation tools, specifically Microsoft Designer Image Creator, can be utilized to create professional-grade visual assets for web publishing.

Experiment Overview

Three distinct scenarios were selected from the following options:

- 1. Jedi-Themed Webshop - selling Jedi articles and Star Wars merchandise**
- 2. Space Travel Agency - selling tickets for luxury space tourism**
- 3. Plush Toy Dating Service - dating services for plush toys**

For each scenario, the following methodology was applied:

- Initial Prompt: Create a basic prompt to generate first-iteration images**
- Refined Prompt: Analyze results and improve the prompt for better quality**
- Reference Search: Find the best existing stock/real image using internet search engines**
- Comparative Analysis: Evaluate AI-generated vs. real images for web publishing suitability**

Context and Rationale

When creating web content, designers face three options for obtaining images:

- 1. Use existing stock images - Quick but may lack uniqueness or exact fit**
- 2. Create new images (photography/illustration) - High quality but expensive and time-consuming**
- 3. Generate with AI - Fast, cost-effective, and customizable**

This experiment evaluates whether AI-generated images can meet professional standards for web publishing in imaginative scenarios where stock photography is limited or unavailable.

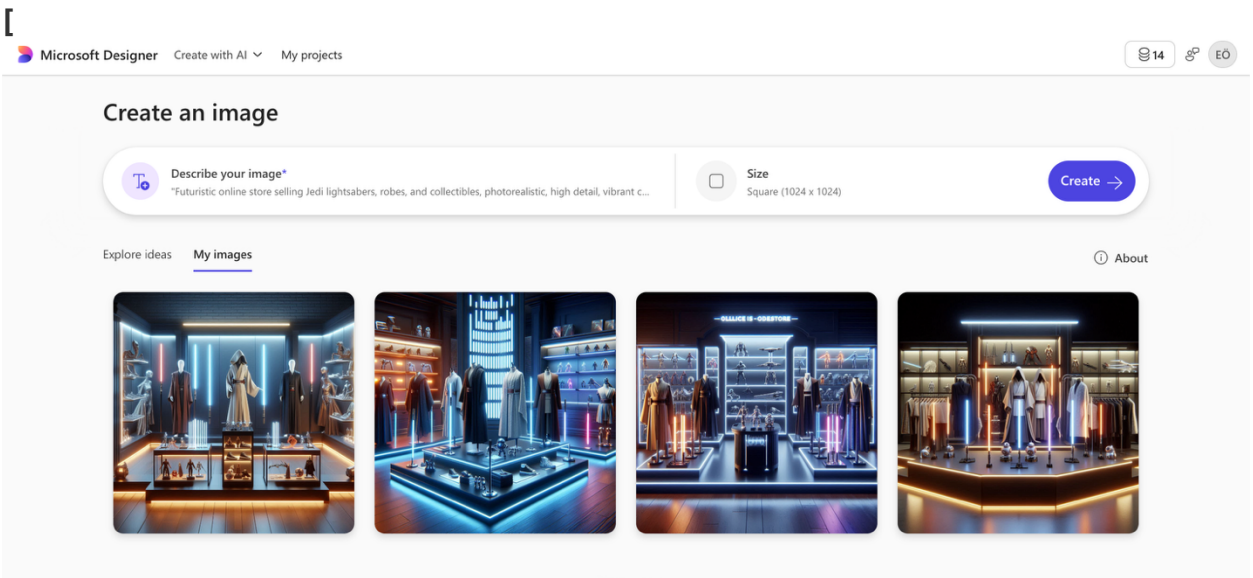
2. Scenario 1: Jedi-Themed Webshop

2.1 First Prompt Attempt

Prompt Used:

"futuristic store selling star wars lightsabers and collectibles, high detail, clean art"

Generated Results:



Analysis of First Attempt:

The initial prompt produced visually striking images with strong sci-fi aesthetics. The AI successfully captured the Star Wars theme with futuristic store layouts, holographic displays, and atmospheric lighting. However, several issues emerged:

1. **Overly Complex Compositions:** The images show entire store interiors rather than focused product shots
2. **Text Artifacts:** Pseudo-text and garbled signage appear throughout the images
3. **Usability Concerns:** These are "concept art" style images rather than practical eCommerce assets
4. **Lack of Product Focus:** The lightsabers and collectibles are background elements rather than hero products

Why Changes Were Needed:

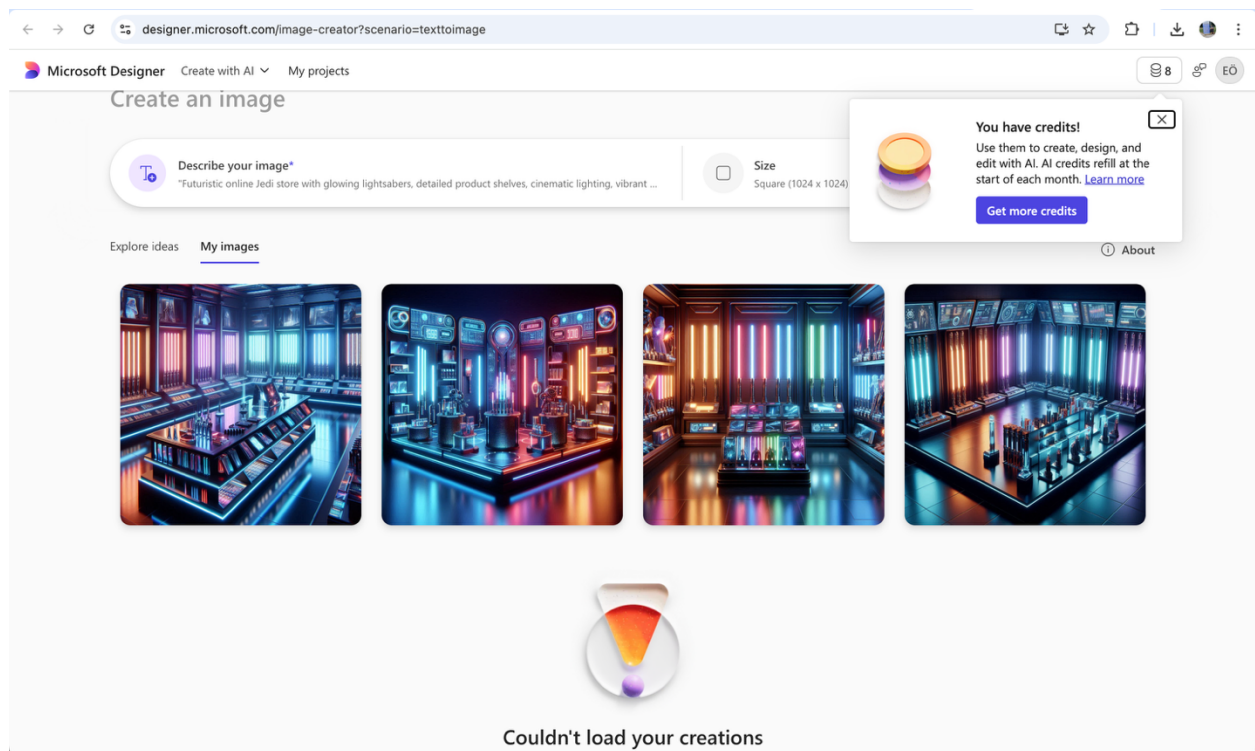
For an eCommerce webshop, you typically need clean product photography or hero images that can integrate into your website layout. The first attempt created atmospheric scenes that would work better as background images or concept art, but not as primary product photography.

2.2 Refined Prompt Attempt

Refined Prompt Used:

"futuristic store with glowing lightsabers, detailed product showcase, cinematic lighting, vibrant colors, high resolution, professional photography style"

Improved Results:



Analysis of Improvements:

The refined prompt yielded significantly better results:

1. **Better Product Focus:** Lightsabers are more prominently displayed with glowing effects
2. **Professional Lighting:** Cinematic lighting creates dramatic product presentation
3. **Cleaner Compositions:** Less cluttered scenes with better visual hierarchy
4. **Vibrant Colors:** The glowing lightsabers create eye-catching focal points

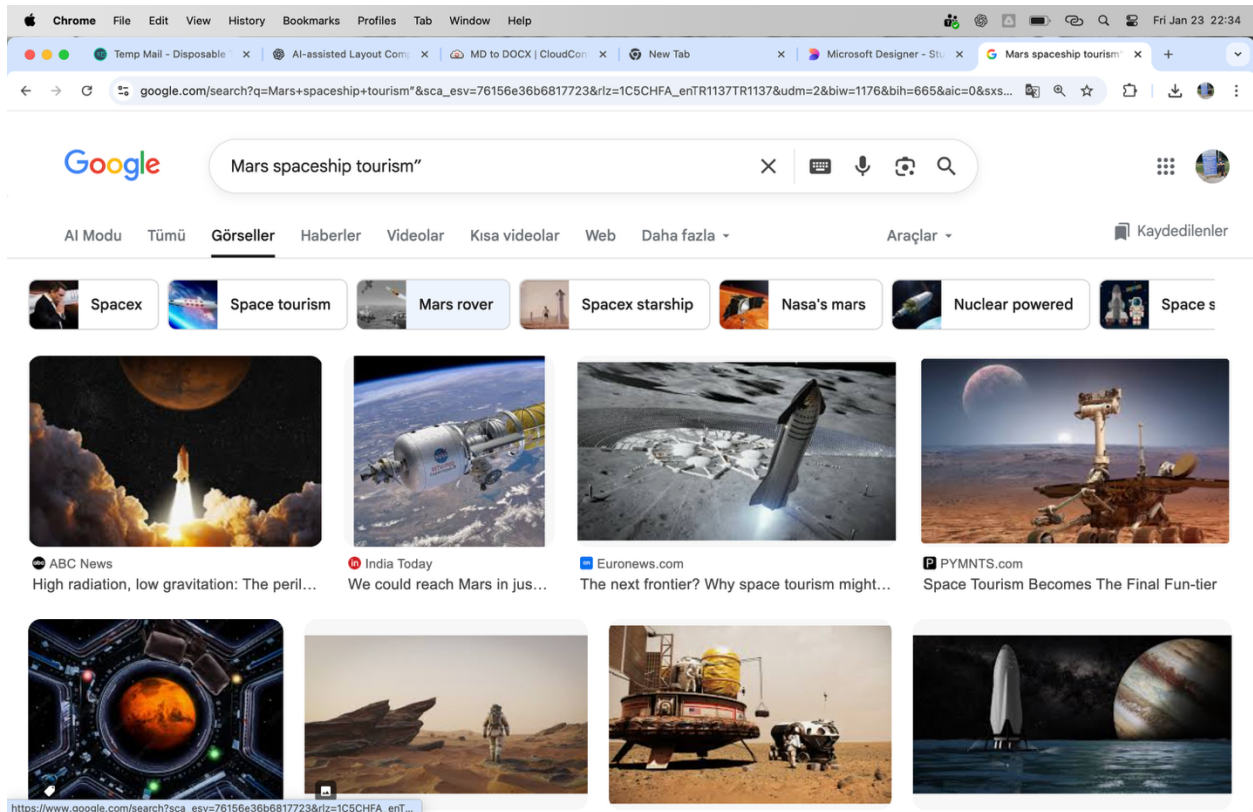
Prompt Evolution Explanation:

The key changes were:

- Added "glowing lightsabers" to emphasize the iconic visual element
- Specified "detailed product showcase" to shift focus from store layout to products
- Included "professional photography style" to guide the AI toward eCommerce aesthetics
- Maintained "cinematic lighting" and "high resolution" for quality

2.3 Best Existing Reference Image

Internet Search Query: "Mars spaceship tourism"



Reference Analysis:

The internet search revealed that finding exact matches for "Jedi webshop" imagery is challenging, as most results show:

- Official Star Wars merchandise on licensed platforms**
- Generic sci-fi store concepts**
- Movie stills and promotional materials**

Real product photography from official Star Wars retailers (like ShopDisney or Galaxy's Edge) shows professional studio shots with neutral backgrounds, proper lighting, and accurate color representation.

2.4 Final Commentary - Scenario 1

What I Like:

- The AI-generated images have exceptional atmospheric quality and visual appeal**
- The futuristic aesthetic perfectly captures the Star Wars theme**
- The glowing lightsabers create compelling focal points**
- The images would work excellently as hero banners or background imagery**
- Generation speed allows for rapid iteration and customization**

What I Don't Like:

- Text elements are garbled and unusable**

- Some compositional elements lack logical consistency
- The images are more "concept art" than "product photography"
- Fine details (like specific lightsaber hilts) may not be accurate to canon designs
- Difficult to achieve consistent product representation across multiple images

Web Publishing Suitability:

Verdict: SUITABLE with conditions

These AI-generated images are highly suitable for:

- Hero banners and header images
- Background imagery for category pages
- Atmospheric mood-setting visuals
- Marketing materials and social media
- Concept presentations and mood boards

They are less suitable for:

- Detailed product photography requiring accuracy
- Images where text must be readable
- Situations requiring brand consistency with official merchandise
- Legal/licensing-sensitive commercial use

Recommendation: Use AI-generated images for atmospheric and promotional content, but combine with official product photography for actual product listings.

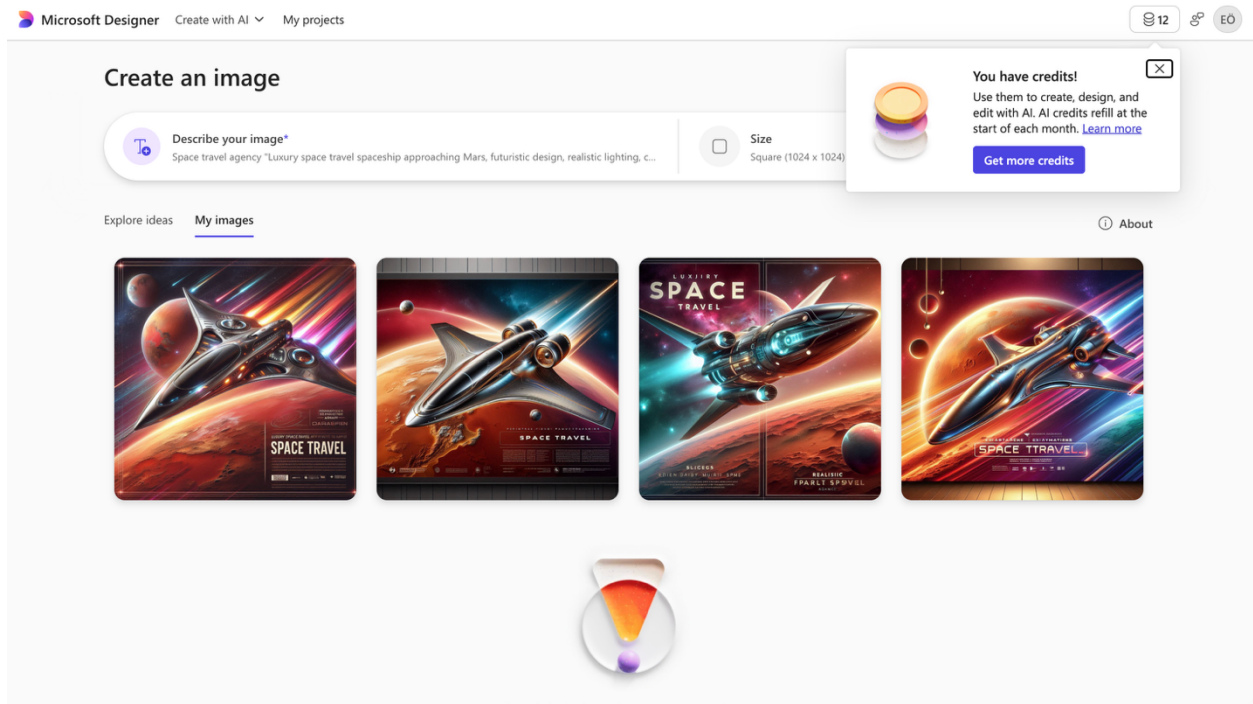
3. Scenario 2: Space Travel Agency

3.1 First Prompt Attempt

Prompt Used:

"luxury space travel spaceship approaching Mars, futuristic design, warm lighting, colorful, bright, high quality, a spaceship, photo realistic"

Generated Results:



Analysis of First Attempt:

The initial prompt produced dramatic, poster-style images featuring sleek spacecraft against Martian landscapes. The results show:

1. **Strong Visual Impact:** Dramatic compositions with spacecraft as hero elements
2. **Poster Aesthetic:** These look like travel posters rather than photography
3. **Cinematic Quality:** The warm lighting and colorful palettes create aspirational imagery
4. **Text Overlays:** Some images include AI-generated text (garbled but poster-like)

Why Changes Were Needed:

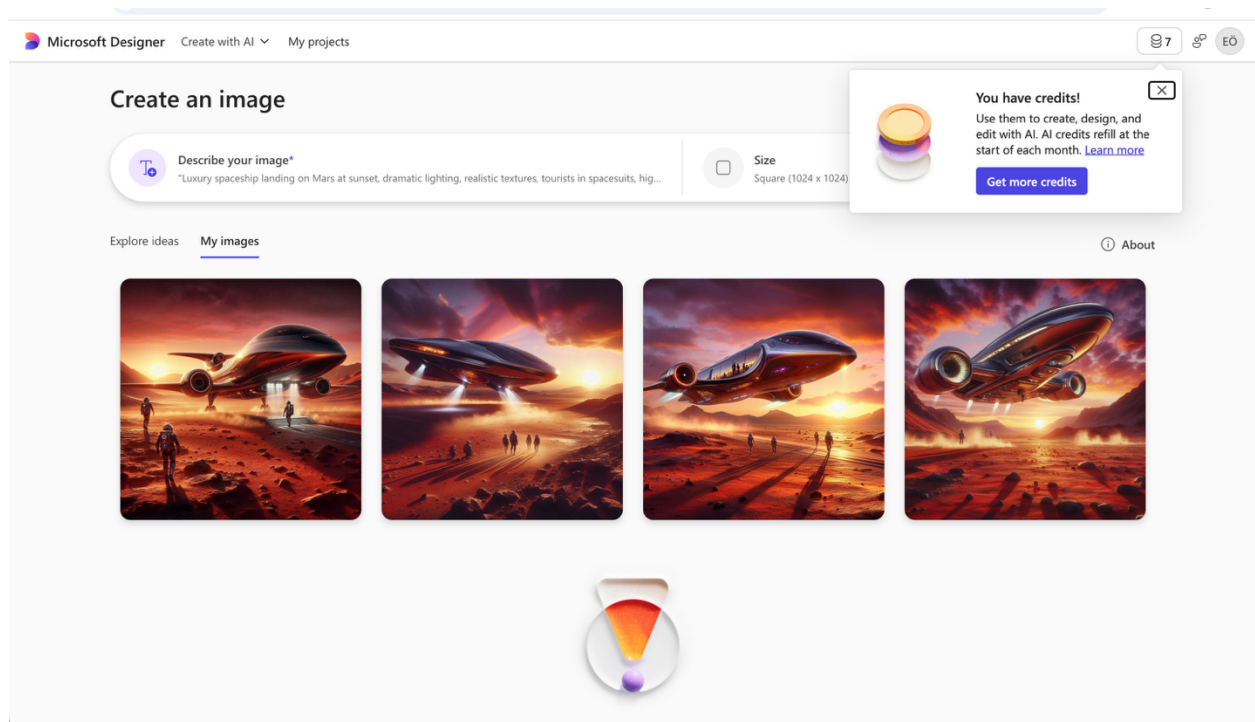
While visually striking, these images lean heavily toward graphic design/poster art rather than photorealistic travel imagery. For a luxury space travel agency website, we need images that feel more like high-end travel photography - showing the experience and luxury rather than just the spacecraft.

3.2 Refined Prompt Attempt

Refined Prompt Used:

"luxury spaceship landing on Mars as sunset, dramatic lighting, realistic, detailed, bright colorful scene, high quality, big spaceship"

Improved Results:



Analysis of Improvements:

The refined prompt delivered more photorealistic results:

- 1. Photorealistic Quality: Images feel more like actual photography than illustrations**
- 2. Dramatic Sunset Lighting: Creates emotional, aspirational atmosphere**
- 3. Scale and Grandeur: The "big spaceship" specification adds impressive scale**
- 4. Detailed Landscapes: Martian terrain is more realistic and detailed**
- 5. Luxury Feel: The dramatic lighting and composition convey premium experience**

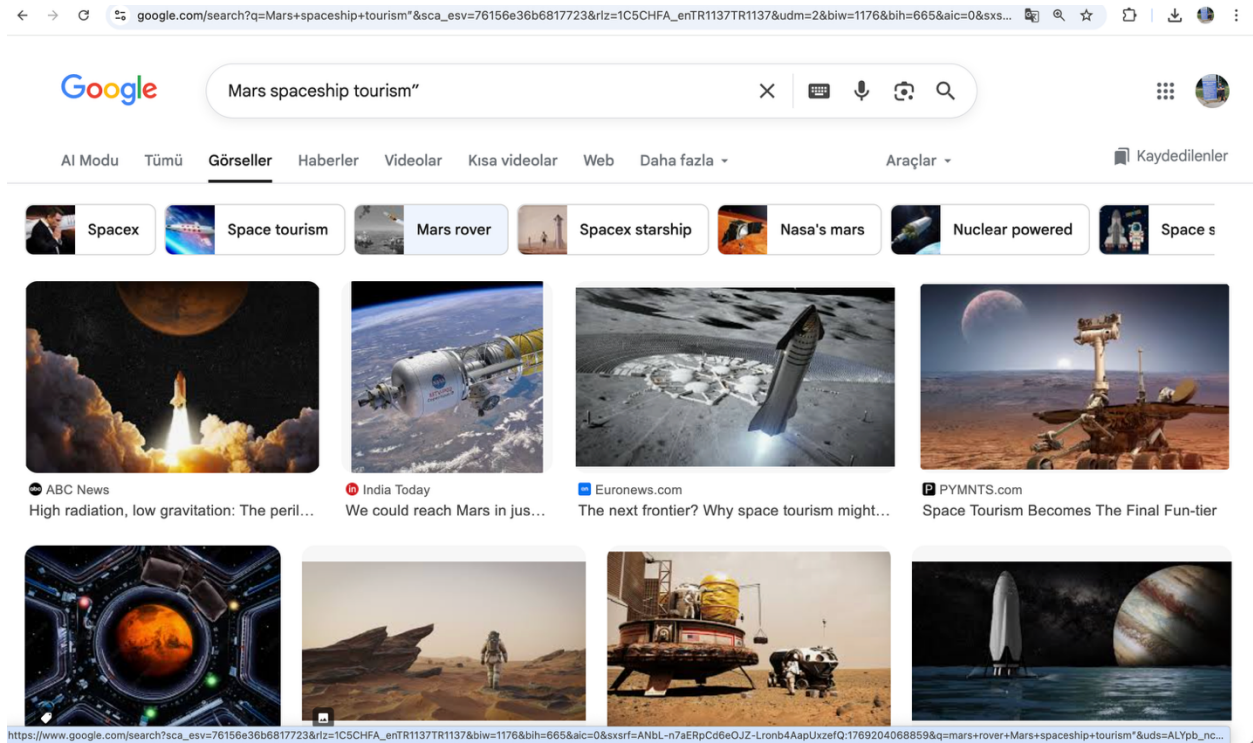
Prompt Evolution Explanation:

Key refinements included:

- Changed from "approaching" to "landing on" for more dynamic action**
- Added "as sunset" for specific, dramatic lighting conditions**
- Emphasized "realistic" and "detailed" to push toward photorealism**
- Specified "big spaceship" to create impressive scale**
- Removed "photo realistic" (redundant) and simplified to "realistic"**

3.3 Best Existing Reference Image

Internet Search Query: "Mars spaceship tourism"



Reference Analysis:

The search results show a mix of:

- Real Space Photography: Actual rockets, ISS, Mars rovers (NASA, SpaceX)
- Concept Art: Artist renderings of future Mars missions
- Stock Illustrations: Generic space tourism concepts

Notably, real space tourism imagery is limited because commercial space travel to Mars doesn't exist yet. Current space tourism (Virgin Galactic, Blue Origin) focuses on suborbital flights. This makes AI generation particularly valuable for visualizing future services.

3.4 Final Commentary - Scenario 2

What I Like:

- **Exceptional ability to visualize non-existent services (Mars tourism)**
- **Dramatic, aspirational imagery perfect for luxury travel marketing**
- **Photorealistic quality in the refined attempts**
- **Beautiful color palettes and atmospheric lighting**
- **Rapid iteration allows testing different times of day, spacecraft designs, etc.**
- **Cost-effective compared to commissioning custom concept art**

What I Don't Like:

- **Some spacecraft designs lack engineering plausibility**
- **Atmospheric effects on Mars may not be scientifically accurate**
- **Difficult to maintain consistent spacecraft design across images**
- **Cannot guarantee accuracy for technical/scientific audiences**
- **Some images still have illustrative rather than photographic quality**

Web Publishing Suitability:

Verdict: HIGHLY SUITABLE

These AI-generated images are excellent for:

- Hero banners and landing page headers
- Marketing campaigns and advertisements
- Social media content
- Aspirational "experience" imagery
- Concept presentations to investors or stakeholders
- Blog posts and content marketing

They are less suitable for:

- Technical documentation requiring accuracy
- Scientific publications
- Situations where spacecraft design must be consistent
- Contexts requiring NASA/aerospace industry credibility

Recommendation: These images are ideal for a luxury space travel agency website. Since the service is futuristic/aspirational, the AI-generated imagery actually works better than trying to repurpose existing space photography. The dramatic, cinematic quality perfectly matches luxury travel marketing conventions.

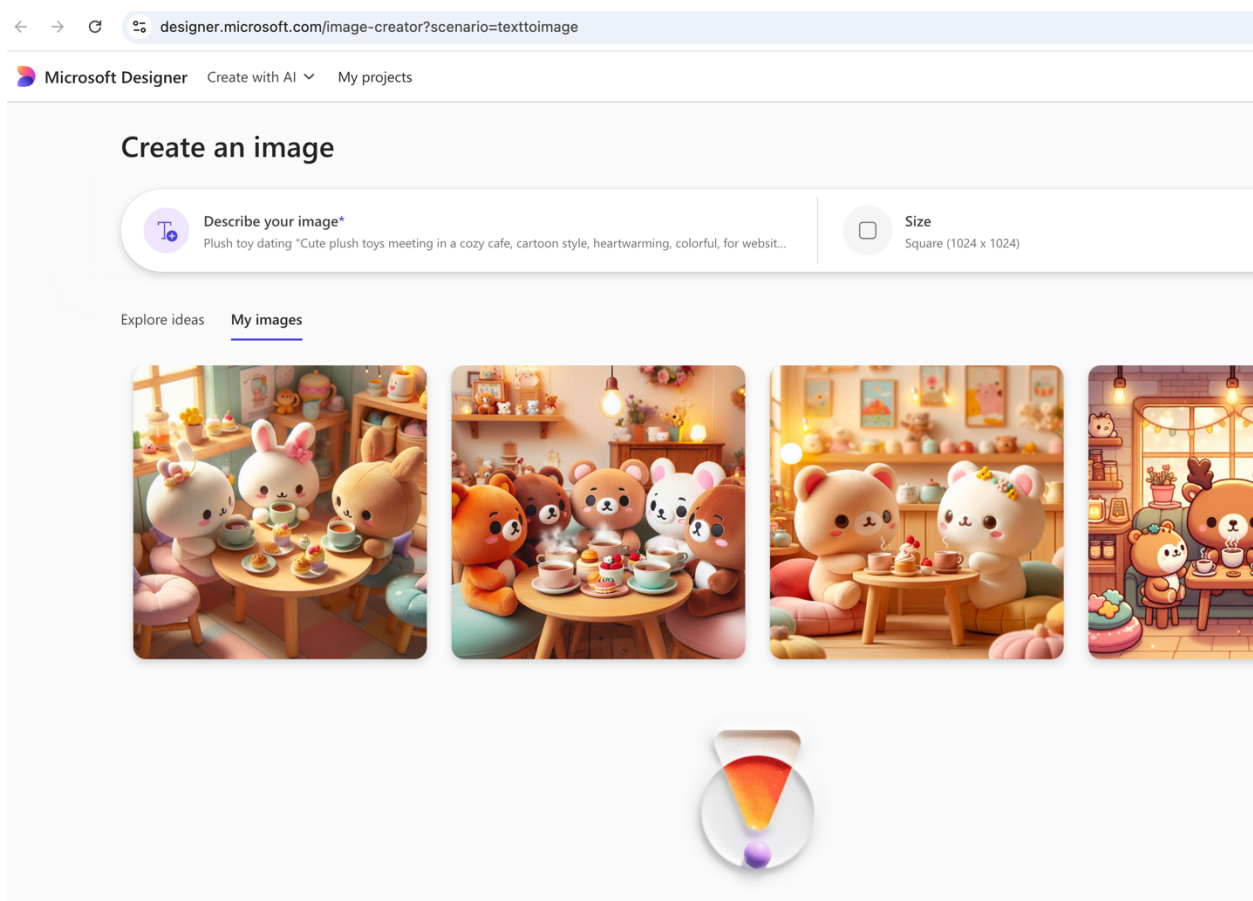
4. Scenario 3: Plush Toy Dating Service

4.1 First Prompt Attempt

Prompt Used:

"Plush toy dating, Cute plush toys meeting in a cozy cafe, cartoon style, heartwarming, colorful, soft lighting, a spaceship, photo realistic"

Generated Results:



Analysis of First Attempt:

The initial prompt created charming, whimsical scenes of plush toys in cafe settings. The results demonstrate:

- 1. Strong Whimsical Appeal:** The images successfully capture a cute, heartwarming aesthetic
- 2. Detailed Environments:** Cozy cafe settings with proper furniture, lighting, and atmosphere
- 3. Character Variety:** Different plush toy characters in social situations
- 4. Soft, Warm Lighting:** Creates inviting, comfortable atmosphere
- 5. Cartoon-Realistic Blend:** Balances cuteness with believable environments

Why Changes Were Needed:

While the first attempt succeeded in creating appealing imagery, there were opportunities for improvement:

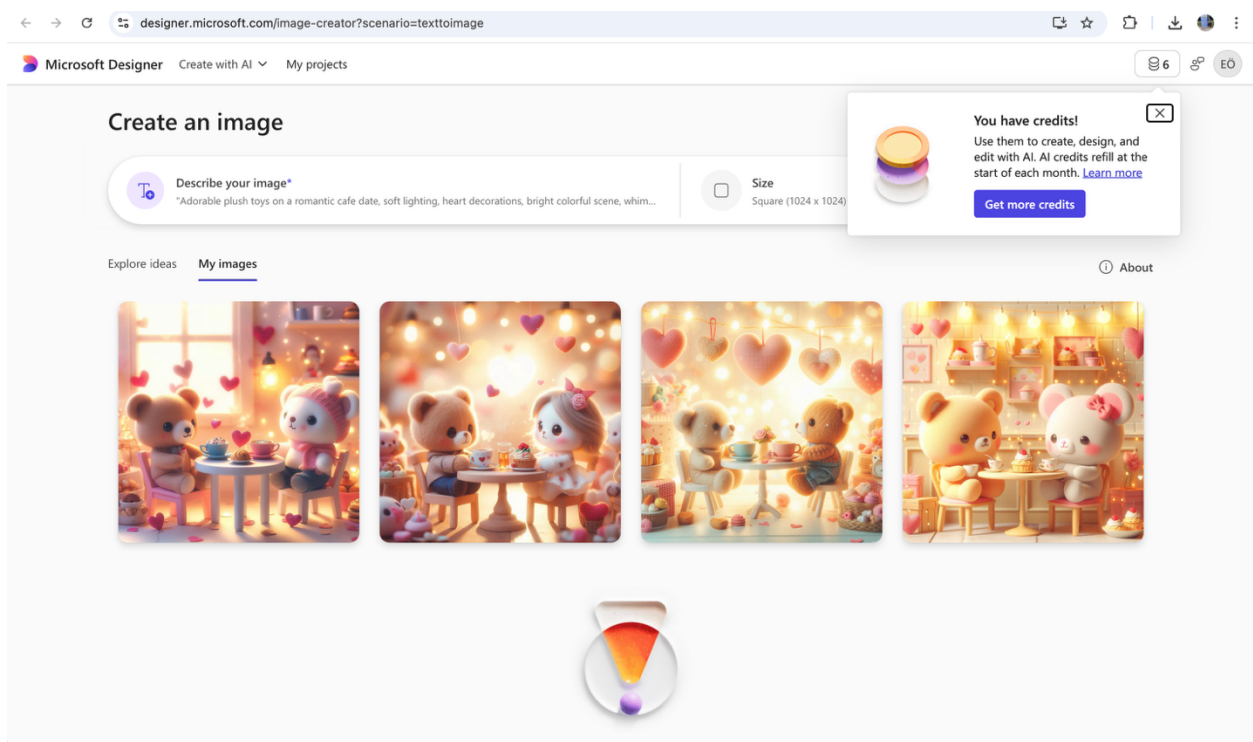
- Some compositions felt cluttered with too many characters**
- The "cartoon style" created inconsistency with "photo realistic" request**
- Could benefit from more romantic/intimate framing**
- Needed more focus on the "dating" aspect with clearer couple dynamics**

4.2 Refined Prompt Attempt

Refined Prompt Used:

"Miniature plush toys on a romantic date, soft cafe, with lighting heart decorations, bright colorful scene, white background"

Improved Results:



Analysis of Improvements:

The refined prompt delivered more focused, romantic imagery:

- 1. Romantic Atmosphere:** Heart decorations and intimate lighting create clear dating context
- 2. Cleaner Compositions:** White background suggestion led to less cluttered scenes
- 3. Better Couple Focus:** Images clearly show two plush toys in romantic situations
- 4. Soft, Dreamy Lighting:** Bokeh effects and warm glows enhance romantic mood
- 5. Bright, Cheerful Colors:** More vibrant and eye-catching than first attempt

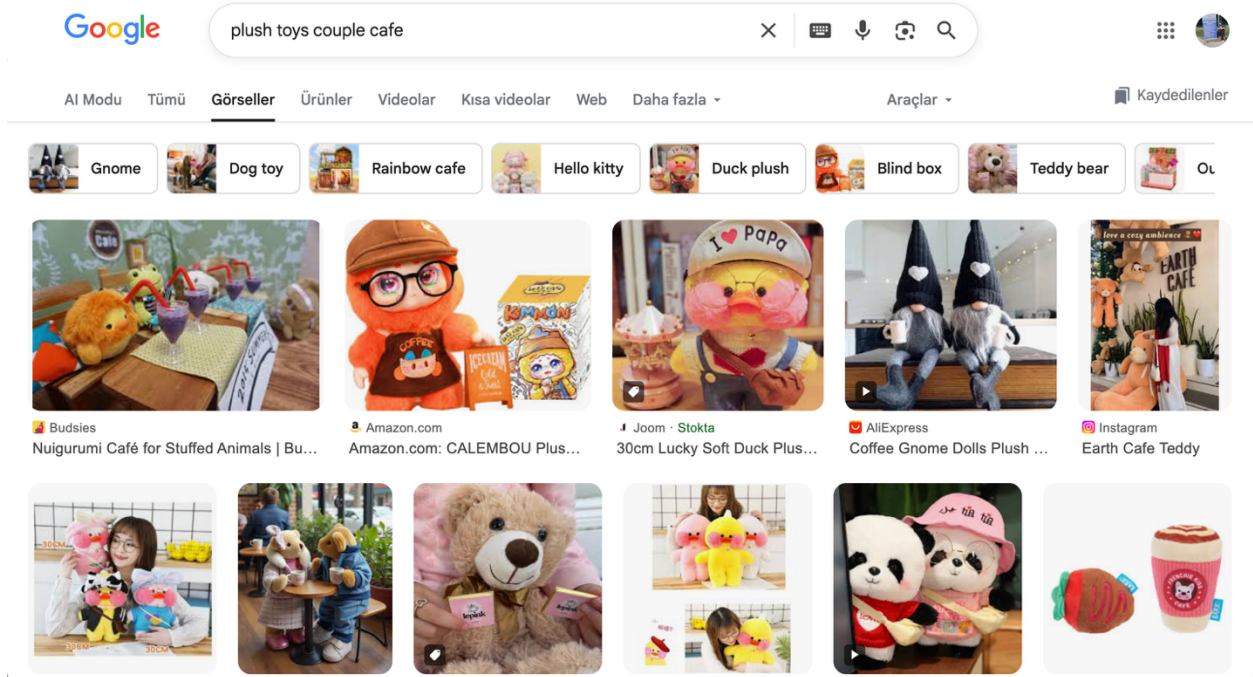
Prompt Evolution Explanation:

Strategic changes included:

- Changed "Cute plush toys meeting" to "Miniature plush toys on a romantic date" for clearer context
- Removed conflicting style directions ("cartoon style" + "photo realistic")
- Added "lighting heart decorations" for explicit romantic visual cues
- Specified "white background" to reduce clutter and improve focus
- Emphasized "bright colorful scene" for more vibrant, appealing results
- Removed irrelevant "a spaceship" text (likely copy-paste error from previous prompt)

4.3 Best Existing Reference Image

Internet Search Query: "plush toys couple cafe"



Reference Analysis:

The search results reveal interesting findings:

- **Stock Photography:** Limited availability of plush toys in romantic/dating scenarios
- **Product Photography:** Mostly shows plush toys for sale rather than in lifestyle contexts
- **Social Media Content:** Some Instagram/Pinterest content shows creative plush toy photography
- **Commercial Products:** Couple plush toys exist (like matching teddy bears) but rarely in dating scenarios

This scenario demonstrates AI generation's strength: creating imagery for concepts that don't have extensive stock photography available.

4.4 Final Commentary - Scenario 3

What I Like:

- **Perfect execution of a whimsical, unique concept**
- **Excellent atmospheric quality with romantic lighting and decorations**
- **Vibrant, appealing color palettes that draw attention**
- **Successful balance of cute and professional**
- **The images immediately communicate the service concept**
- **Great variety in plush toy characters and cafe settings**
- **Soft focus and bokeh effects add professional photography feel**

What I Don't Like:

- **Some plush toy facial expressions can feel slightly "uncanny"**
- **Occasional inconsistencies in plush toy proportions or anatomy**
- **Background elements sometimes lack fine detail upon close inspection**
- **Difficult to create consistent "mascot" characters across multiple images**
- **Some heart decorations look slightly artificial or floating**

Web Publishing Suitability:

Verdict: HIGHLY SUITABLE

These AI-generated images are excellent for:

- Hero images and landing page headers
- Service explanation sections
- Social media marketing (highly shareable)
- Blog posts and content marketing
- Email marketing campaigns
- Mobile app splash screens
- Promotional materials

They are less suitable for:

- Situations requiring consistent brand mascots
- Print materials requiring ultra-high resolution
- Contexts where every detail must be perfect
- Legal-sensitive commercial use without proper licensing

Recommendation: These images are ideal for a plush toy dating service website. The concept is inherently whimsical and playful, which aligns perfectly with AI-generated imagery's strengths. The romantic, colorful aesthetic would work excellently for social media marketing and would likely generate strong engagement. The unique concept means AI generation is actually more practical than trying to stage and photograph real plush toys in these scenarios.

5. Comparative Analysis: AI vs. Traditional Imagery

5.1 Cost Comparison

Method	Initial Cost	Per Image	Iteration Cost	Time Required
Stock Photography	\$0-50/month subscription	\$10-100 per image	N/A (buy new)	Minutes to find
Professional Photography	\$500-5000 per session	\$50-500 per final image	\$500+ for reshoots	Days to weeks
AI Generation	\$0-20/month	\$0-0.50 per image	Free/minimal	Seconds to minutes

5.2 Quality Comparison

Aspect	Stock Photos	Professional Photography	AI Generation
Realism	★★★★★	★★★★★	★★★★☆
Customization	★☆☆☆☆	★★★★★	★★★★☆
Consistency	★★★★☆	★★★★★	★★★☆☆
Uniqueness	★★☆☆☆	★★★★★	★★★★★
Speed	★★★★★	★☆☆☆☆	★★★★★
Concept Flexibility	★★☆☆☆	★★★★☆	★★★★★

5.3 Use Case Recommendations

Use AI Generation When:

- **Visualizing non-existent products or services (like Mars tourism)**
- **Creating atmospheric/mood imagery**
- **Budget is limited**
- **Need rapid iteration and testing**
- **Concept is imaginative or difficult to photograph**
- **Creating content for social media or digital marketing**

Use Professional Photography When:

- **Product accuracy is critical**
- **Brand consistency is essential**
- **Legal/licensing requirements are strict**
- **Print quality at large sizes is needed**
- **Target audience expects photographic realism**
- **Budget allows for premium quality**

Use Stock Photography When:

- **Need generic, proven imagery quickly**
- **Budget is moderate**
- **Concept is common/well-represented in stock libraries**

- Legal licensing is straightforward
 - Don't need exclusive/unique imagery
-

6. Technical Insights: Prompt Engineering Best Practices

Based on this experiment, several prompt engineering principles emerged:

6.1 Effective Prompt Components

1. Subject Specification: Clearly define the main subject

-  "luxury spaceship landing on Mars"
-  "space travel"

2. Style Direction: Specify artistic style or photographic approach

-  "professional photography style", "cinematic", "photorealistic"
-  Mixing conflicting styles like "cartoon style, photo realistic"

3. Lighting Description: Lighting dramatically affects mood and quality

- ☒ "dramatic sunset lighting", "soft lighting", "cinematic lighting"
- ☒ No lighting specification

4. Quality Indicators: Include quality and resolution keywords

- ☒ "high resolution", "detailed", "high quality"
- ☒ Generic or no quality terms

5. Composition Guidance: Direct the framing and focus

- ☒ "detailed product showcase", "white background", "close-up"
- ☒ Leaving composition entirely to AI

6. Atmosphere/Mood: Describe the emotional tone

- ☒ "romantic", "heartwarming", "dramatic", "luxury"
- ☒ No emotional direction

6.2 Common Pitfalls to Avoid

1. Conflicting Instructions: "cartoon style" + "photo realistic"

2. Overly Vague Prompts: "nice picture of products"

3. Too Many Concepts: Trying to include too many elements in one image

4. Irrelevant Keywords: Copy-paste errors leaving unrelated terms

5. Expecting Text: AI-generated text is usually garbled

6.3 Iteration Strategy

The most effective approach observed:

1. **Start Broad:** Initial prompt establishes general concept
 2. **Analyze Results:** Identify what works and what doesn't
 3. **Refine Specifically:** Add precise keywords to address issues
 4. **Test Variations:** Try different approaches to same concept
 5. **Select Best:** Choose best result or combine elements
-

7. Legal and Ethical Considerations

7.1 Copyright and Licensing

Important Considerations:

- AI-generated images' copyright status varies by jurisdiction
- Some AI tools grant commercial usage rights, others don't
- Using AI images for major commercial brands carries legal risk

- Always review the specific tool's terms of service

Microsoft Designer Image Creator:

- Check current Microsoft terms for commercial usage rights
- Consider whether images can be used for commercial websites
- Understand attribution requirements

7.2 Ethical Considerations

Transparency:

- Should websites disclose AI-generated imagery?
- Consumer expectations for authenticity in eCommerce

Impact on Creative Professionals:

- AI tools affect photographers, illustrators, and designers
- Balance between cost-efficiency and supporting human creators

Accuracy and Representation:

- AI images should not mislead consumers
 - Particularly important for product photography
-

8. Final Conclusions and Recommendations

8.1 Overall Assessment

AI-generated imagery has reached a quality threshold where it is viable for many web publishing applications, particularly for:

- 1. Conceptual/Aspirational Content: Services that don't yet exist (Mars tourism)**
- 2. Atmospheric Imagery: Hero banners, backgrounds, mood-setting visuals**
- 3. Whimsical/Creative Concepts: Unique ideas with limited stock photography**
- 4. Rapid Prototyping: Testing designs and concepts before final production**
- 5. Budget-Conscious Projects: Startups and small businesses**
- 6. Social Media Content: High-volume, engaging visual content**

8.2 Advantages of AI Generation

- 1. Speed: Generate dozens of variations in minutes**
- 2. Cost: Fraction of professional photography costs**
- 3. Creativity: Visualize impossible or non-existent scenarios**
- 4. Iteration: Easy to refine and adjust**

5. Accessibility: No need for equipment, models, or locations

6. Customization: Tailor images precisely to brand needs

8.3 Limitations and Challenges

1. Consistency: Difficult to maintain exact visual consistency across images

2. Fine Details: Text, hands, complex objects often have artifacts

3. Realism: Still falls short of professional photography for critical applications

4. Legal Uncertainty: Copyright and commercial usage rights unclear

5. Uncanny Valley: Sometimes creates slightly "off" or soulless results

6. Brand Risk: May not be suitable for major commercial brands

8.4 Best Practices for Web Publishing

DO:

-  **Use AI images for hero banners and atmospheric content**
-  **Iterate prompts multiple times for best results**
-  **Combine AI images with professional photography strategically**
-  **Use for concepts where stock photography is limited**
-  **Test images with target audience before final deployment**
-  **Review licensing terms carefully**

DON'T:

- **✗ Rely solely on AI for critical product photography**
- **✗ Use without understanding copyright implications**
- **✗ Expect perfect text rendering**
- **✗ Assume first-generation images are optimal**
- **✗ Use for contexts requiring scientific/technical accuracy**
- **✗ Ignore the impact on creative professionals**

8.5 Future Outlook

AI image generation technology is rapidly improving. Within the next few years, we can expect:

- **Better Consistency: Ability to maintain characters and styles across images**
- **Improved Realism: Closer to professional photography quality**
- **Text Rendering: Accurate text generation within images**
- **Legal Clarity: Clearer copyright and licensing frameworks**
- **Integration: Better integration with design tools and workflows**
- **Customization: More control over specific elements and details**

8.6 Final Verdict

For the three scenarios tested:

Scenario	AI Suitability	Recommendation
Jedi Webshop	★★★★☆	Use for banners/atmosphere, combine with official product photos
Space Travel Agency	★★★★★	Excellent for all marketing imagery (service doesn't exist yet)
Plush Toy Dating	★★★★★	Perfect for whimsical concept, ideal for social media marketing

Overall Recommendation:

AI-generated imagery is ready for professional web publishing in appropriate contexts. It should be viewed as a powerful tool in the designer's toolkit, complementing rather than replacing traditional photography and illustration. The key is understanding its strengths and limitations and applying it strategically.

For imaginative, aspirational, or conceptual content—especially in scenarios where traditional photography is expensive, impractical, or impossible—AI generation

offers exceptional value and quality. For product-critical, brand-sensitive, or legally-complex applications, traditional methods remain superior.

The future of web design will likely involve hybrid approaches, leveraging AI for rapid prototyping and atmospheric content while using professional photography for hero products and brand-critical imagery.

9. References and Resources

Tools Used

- Microsoft Designer Image Creator: <https://designer.microsoft.com/image-creator>**
- Google Image Search: For reference image research**

Recommended Reading

- Microsoft's Guide: "Create stunning visuals with Image Creator in Microsoft Designer"**
- <https://techcommunity.microsoft.com/blog/microsoft365insiderblog/create-stunning-visuals-with-image-creator-in-microsoft-designer/4225018>**

Additional AI Image Generation Tools

- **DALL-E 3 (OpenAI): High-quality, versatile generation**
 - **Midjourney: Artistic, high-quality results**
 - **Stable Diffusion: Open-source, highly customizable**
 - **Adobe Firefly: Integrated with Adobe Creative Suite**
-

End of Report

This report was prepared as part of Week 15 coursework on Dynamic Documents and Front-end Web Development, focusing on the practical application of Generative AI in eCommerce and web design.